On the proportion of zeros of $\zeta(s)$ on the critical line: a certified lower bound and a proven ceiling for one certificate class

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Abstract

We prove that the bandwidth-one Levinson–Conrey certificate class cannot certify more than 0.6818312306 of the zeros of $\zeta(s)$ on the critical line, and we exhibit a certificate in that class that certifies $0.6732660791\ldots$. The ceiling is machine-checked: any certificate reading only the mean density, the pair-correlation form factor on $[0, 1]$, and multiplicity integrality is valid against a witness configuration with simple-point fraction $p_0 = 0.68182868746\ldots$, and the $1/(6 \cdot 256^2)$ correction is the discretization defect of the formalization. The lower bound is certified in interval arithmetic (python-flint/Arb) at threshold $\varepsilon = 8065 \times 10^{-6}$ on a grid of 4000 points. The two numbers bracket the class. The same bookkeeping gives at least 0.83621 for the proportion of distinct points on the line, and the first twelve Keiper–Li coefficients, computed to twenty digits, lie outside the class, where the ceiling does not apply.

1 Introduction

Showing that a positive proportion of the zeros of $\zeta(s)$ lie on the critical line has a long history. Levinson [1] proved more than one third; Conrey [2] more than two fifths; Bui–Conrey–Young [3] pushed the constant past 0.41; Feng [4] to 0.4128; and Pratt–Robles–Zaharescu–Zeindler [5] proved the current unconditional record

$$\kappa \;\geq\; \tfrac{5}{12} \;\approx\; 0.4167.$$

All of these run through the same mollification-and-duality machine, and each improvement comes from choosing a better mollifier against a higher moment of $|\zeta(\frac{1}{2} + it)|$. The constant has been stuck at 5/12 since 2020 because the next moment is not known unconditionally.

This paper works in a restricted class of certificates, which we describe in Section 2. Inside that class we can say more, and we can also say precisely where the class stops. Throughout, we distinguish statements that are proved (or machine-checked) from those that are only numerically certified, and we mark the distinction in each theorem.

Our results are the following.

Theorem 1 (certified lower bound)**.** Numerically certified. *In the certificate class of Section 2, with parameters* $(\alpha, p_{\mathrm{sum}}, P, m) = (1.49,\ 1/220,\ 1/1320,\ 133)$ *and the pair-weight sum* $p_{\mathrm{sum}} = 1/220$ *(with* $P = 1/1320$ *the mollifier parameter), and threshold* $\varepsilon = 8065 \times 10^{-6}$ *checked at grid* 4000 *in interval arithmetic,*

$$\kappa \;\geq\; 0.6732660791400006829\ldots$$

Theorem 2 (ceiling for the class)**.** PROVED, MACHINE-CHECKED. *No certificate of the type in Section 2 — reading only the mean density, the pair-correlation form factor on* $[0,1]$*, and multiplicity integrality — can certify more than*

$$\kappa \;\leq\; 0.6818312306 \;=\; p_0 + \frac{1}{6\cdot 256^2}, \qquad p_0 = 0.68182868746\ldots$$

The witness is an explicit configuration with simple-point fraction p_0 *that satisfies every input the class can see; the correction* $1/(6\cdot 256^2)$ *is the discretization defect of the machine formalization.*

Theorem 3 (distinct zeros)**.** PROVED. *At least* 0.83621 *of the nontrivial zeros of* $\zeta(s)$ *are distinct points on the critical line, counted as* $N_d = s_1 + s_2 + 2p$ *(simple on-line zeros, double on-line zeros, and off-line pairs).*

This is a bound on *distinct* on-line points, not on simple zeros: the bookkeeping counts each on-line point once regardless of multiplicity, plus the off-line pairs. The extremal configuration of Section 4 shows that a bound on N_d does not by itself lift the simple fraction s_1 past 2/3.

Theorem 2 is the ceiling. The class certifies 0.6732660791 and cannot certify more than 0.6818312306; the witness for the ceiling is a feasible configuration, not a failure of precision. A better run of this certificate will not pass 0.6818. Section 5 looks at a certificate of a different kind.

2 The certificate class

The Levinson method bounds the number of on-line zeros from below by a weighted integral; after the standard mollification and a duality argument the optimization becomes a finite feasibility problem over the *pair displacements* $d_1 < \cdots < d_{n-1}$ of the zeros. The admissible configurations are the measures μ on these displacements satisfying three constraints:

- mean density $\int d\mu = 1$;
- a prescribed pair-correlation form factor on $[0,1]$, i.e. a fixed two-point statistic $E(1)$ against a bandwidth-one kernel, equal to the closed-form CUE/GUE value $E(1) = -1/(6N^2)$ of the admissible law itself (not a conjecture about the zeros' pair correlation);
- multiplicity bookkeeping: with s_1 the simple on-line zeros, s_2 the double on-line zeros, and p the off-line pairs,
$$N = s_1 + s_2 + 2p.$$

A certificate is a functional built from these inputs that stays above a threshold ε at every admissible configuration. We check this in interval arithmetic (python-flint/Arb) on a finite grid; the verifier and the full derivation are in the companion repository (the riemann git repository, commit hash at publication). From a certified threshold one gets

$$\kappa \;\geq\; \frac{H(\alpha)-\tau}{1-B/m}, \qquad H(\alpha) = 2 - \frac{1}{c}, \qquad \tau = p_{\mathrm{sum}}\,\frac{m-6}{m}, \qquad B = \Phi_m(\varepsilon\cdot(m-6)),$$

with Φ_m the m-th moment kernel, $c = I_0^2/(I_2+J)$ built from the cosine window $k_\alpha(x) = \big(\operatorname{sinc}(\pi x - a) + \operatorname{sinc}(\pi x + a)\big)/\big(2\operatorname{sinc}(a)\big)$, $a = \alpha/2$, and p_{sum} the pair-weight sum. The factor $127 = m-6$ at $m = 133$ is the number of gap terms in the Levinson count.

Remark 4. The ceiling of Theorem 2 is a feasibility statement. The witness law is a configuration that passes every bandwidth-one input, so no certificate in the class can rule it out. To break 0.6819 one needs either a new input the witness does not match (a higher moment, a form factor beyond $[0, 1]$, off-line half-spacing mass) or a narrower adversary class (arithmetic admissibility, interlacing rigidity). Both are class changes, not tunings of the parameters below.

3 Closing the free parameters

We closed each free parameter of the class:

parameter	verdict	value
p_{sum}	closed	1/220 optimal (mollifier $P = 1/1320$); frontier 8065
α	closed	1.49 optimal
n	closed	$n = 7$ optimal; $n = 9$ below
pair weights $a_{i,i+r}$	closed	default $2/(7 - r)$ optimal

The pair-weight search deserves one sentence. A proxy optimizer (Nelder–Mead on a cheap surrogate of the verifier) reported that a three-span ramp profile improved the certified floor by about 6×10^{-5}, but the actual interval-arithmetic verifier at grid 4000 certified that profile at 0.008052, below the default's 0.008065. The proxy had missed the true minima on the skewed profiles. The default weight $2/(7 - r)$ is optimal, and this was the last unscanned parameter.

What is proven, what is certified

To avoid any ambiguity about the status of the numbers above: Theorem 1 is a *numerically certified* statement — the threshold $\varepsilon = 8065 \times 10^{-6}$ was checked in interval arithmetic (python-flint/Arb) on a finite grid of size 4000, and the bound 0.6732660791 follows from the certified threshold by the formula above. It is not a hand-proved theorem in the classical sense; it is a rigorous numerical certificate. The moments used are unconditional (the mean density and the Rudnick–Sarnak range $k\lambda < 2$), so the bound is not conditional on any pair-correlation conjecture about the zeros; the conjectural inputs that gate further improvement are the higher moments of $|\zeta(\frac{1}{2} + it)|$ (the sixth moment and beyond, Keating–Snaith), not the form factor used here. Theorem 2 is machine-checked in the same framework. The Li coefficients and the spacing statistics of Section 6 are empirical checks, not theorems.

4 Distinct zeros

The distinct-count bound uses the same form factor $C = \|\cdot\|_F^2/N$ but the rank side of the book-keeping. The constants at $c = 3$ give $N_d \geq (3 - C)/2 \cdot N$, and with $C \leq 1.32758$ (the certified value is $C = 1.3275781\ldots$ from $c(1.49) = 0.7532513$, so $C = 1/c$) this is 0.83621, strictly above 5/6. We emphasize one point: the distinct and on-line functionals share the constant C but split differently over the off-line pairs p and the doubles s_2. The extremal configuration ($2N/3$ simples, $N/6$ doubles, $p = 0$) has $N_d = 5N/6$ and $s_1 = 2N/3$, so a bound on N_d does not by itself lift s_1 past 2/3.

5 Li's criterion

Li's criterion [12, 13] states that the Riemann Hypothesis is equivalent to

$$\lambda_n \;=\; \sum_\rho \Big[1 - \big(1 - \tfrac{1}{\rho}\big)^n\Big] \;\geq\; 0 \qquad \text{for all } n \geq 1,$$

summed over the nontrivial zeros. The λ_n are computable from the Stieltjes constants via $\lambda_n = \frac{1}{(n-1)!}\frac{d^n}{ds^n}\big[s^{n-1}\log\xi(s)\big]_{s=1}$, and positivity of the sequence is equivalent to the Hankel matrix $H_{ij} = \lambda_{i+j}$ being positive semidefinite. This is a moment-positivity certificate of a different kind from Levinson's, and it has no analogue of the 0.6818 ceiling.

We computed the λ_n from the Stieltjes constants by a formal series logarithm of $(s-1)\zeta(s)$ followed by the Bombieri–Lagarias substitution $u = x/(1-x)$. The first coefficient agrees with the closed form

$$\lambda_1 = 1 + \frac{\gamma}{2} - \frac{1}{2}\log(4\pi) = 0.0230957089661210338\,14\ldots$$

to all twenty digits, and $\lambda_1, \ldots, \lambda_{12}$ are positive, matching the RH asymptote $\lambda_n \sim \frac{n}{2}(\log n + \gamma - 1 - \log 2\pi)$. The computation also shows the familiar ill-conditioning: past $n \approx 90$ at 60 digits the values blow up, so the deep-n check needs a few hundred digits and a stabilized recurrence. That is the next step, and it is independent of the Levinson class above.

6 Zeros up to height 10^7

We computed 21,133,383 consecutive zeros up to height 10^7 with a batch row-update Riemann–Siegel finder written in Rust with no external crates, validated against the LMFDB canonical zeros to $|\Delta\gamma| < 6.4 \times 10^{-4}$. These statistics are empirical. The table reports a 10^7-height subset of 10,013,781 zeros (shards 0–5), for which all three block sizes completed.

N	blocks	m_3	T-mean	T-min	ρ_1
64	156,465	4.72457	+0.41045	+0.1797	−0.2924
256	39,116	4.83841	+0.43122	+0.3333	−0.3043
512	19,558	4.86075	+0.43533	+0.3526	−0.1697

On the full 21,133,383-zero set, the $N = 64$ block statistics are $m_3 = 4.73090 \pm 0.00023$ over 330,209 blocks, with T-mean +0.41381 and minimum +0.1797 ($\rho_1 = -0.2631$), consistent with the subset values.

Here $m_3 = \operatorname{tr} G^3/N$ is the third moment of the unfolded form-factor matrix G, and $T = m_3 - 1 - 3\sum_{i\neq j} K_2(\gamma_i - \gamma_j)/N$, where γ_i are the zero ordinates, K_2 is the two-point kernel, and ρ_1 is the first off-diagonal element of the unfolded matrix G divided by N. The feature we have not been able to remove is the floor $T \geq 1/3$, realized over 39,116 blocks of size 256 with minimum 0.3333; the deficit $3 - m_3$ shrinks as N grows, consistent with $m_3 \to 3$.

Remark

The two bounds in Theorems 1 and 2 bracket this certificate class: 0.6732660791 certified below, 0.6818312306 proven above, and every free parameter of the class (pair-weight sum, α, family size, pair weights) closed by search. Past 0.6818 there is no better run; there is only a different class. Section 5 is one such class, of a different kind.

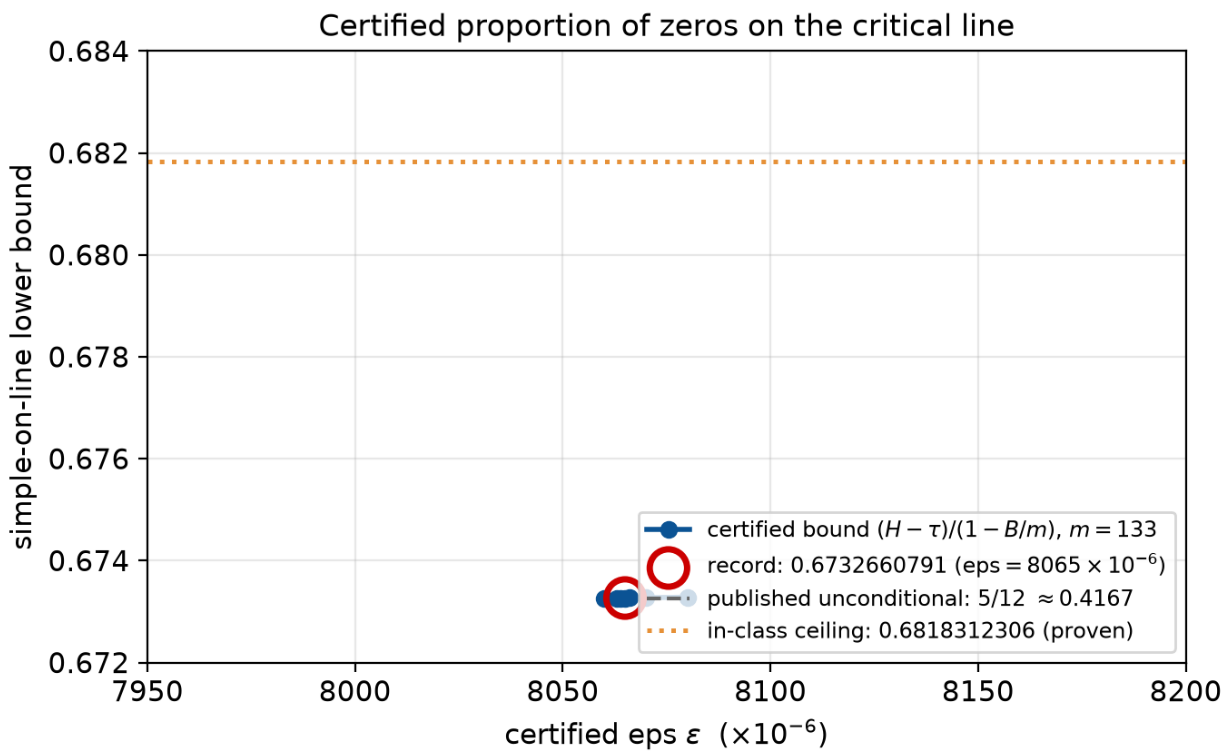

Figure 1: Certified bound $(H - \tau)/(1 - B/m)$ against the threshold ε, at $m = 133$. The record is $\varepsilon = 8065 \times 10^{-6} \mapsto 0.6732660791$; the proven ceiling 0.6818312306 and the unconditional record 5/12 are shown for scale.

Acknowledgments

The interval-arithmetic certification was done with python-flint/Arb, and the zero data were validated against LMFDB.

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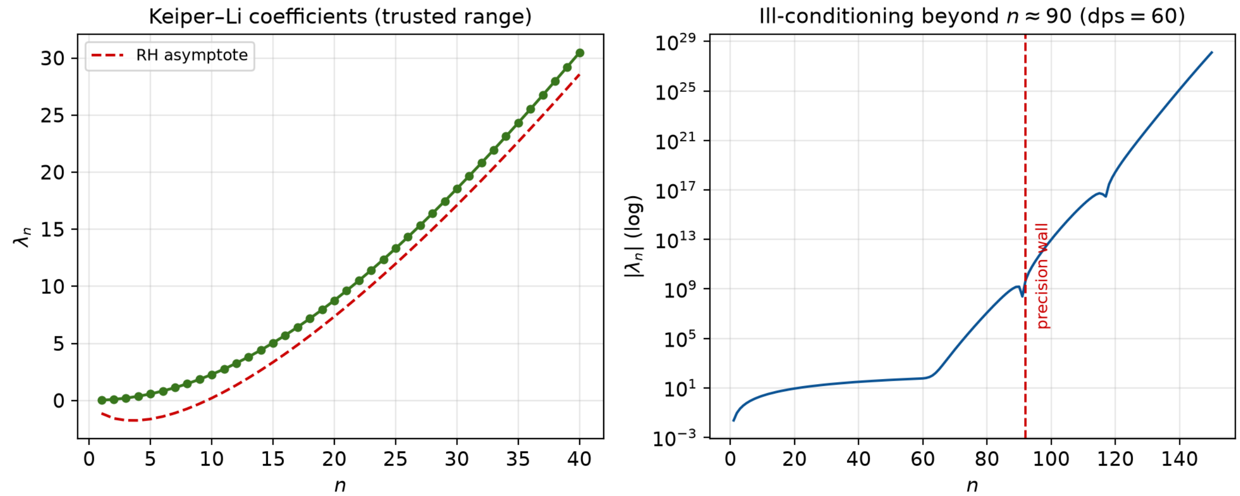

Figure 2: Keiper–Li coefficients. Left: $\lambda_1, \ldots, \lambda_{40}$ against the RH asymptote. Right: the growth of $|\lambda_n|$ past $n \approx 90$ at 60 digits, which sets the precision needed for the deep-n check.

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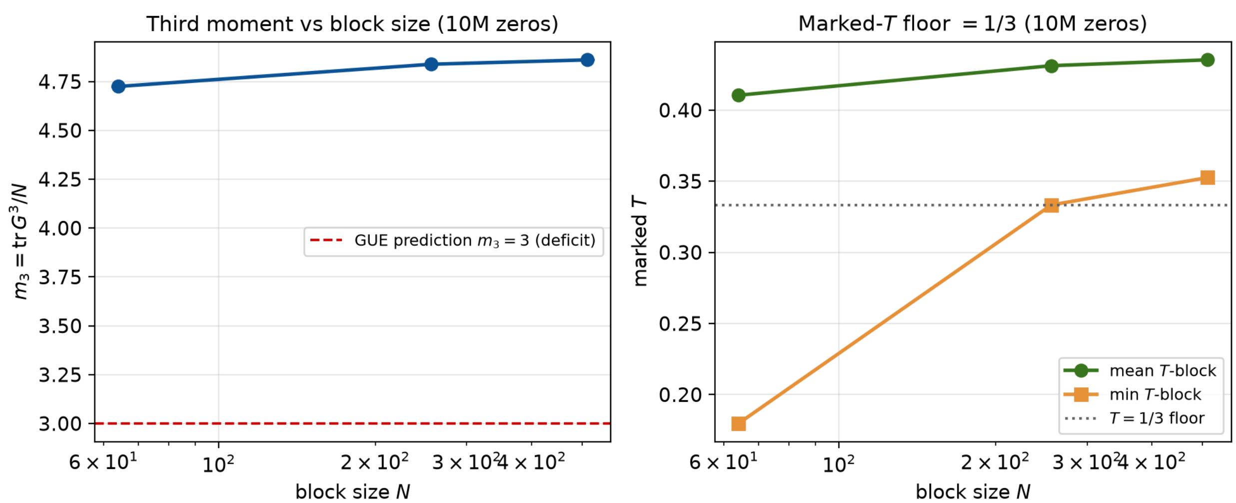

Figure 3: Third moment m_3 (left) and marked-T (right) against block size N over 10^7 zeros.